PARISHRAM 2.0

CLASS 12TH 2026

d & f-Block

CHEMISTRY | LECTURE-2

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TOPICS TO BE COVERED

1 General Properties Of Transition Elements



3d-series / 1st series / 4th period

1st Series										
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Z	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10

4d-series / 2nd series / 5th period

	Yaari	Zara	Nibhana	Mout	Tk	2nd Series	Rote	Rehna	Pdaga	Aaj	Cal
	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	
Z	39	40	41	42	43	44	45	46	47	48	
5s	2	2	1	1	1	1	1	0	1	2	
4d	1	2	4	5	6	7	8	10	10	10	

d-Block Elements

5d-series / 3rd series / 6th period

	La	Haf	ta	Wrna	Remo	Or	Inder	Pitenge	Aur	Hosp jaenge
	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10

3rd Series

Z = 58-71 (lanthanoids)

d-Block Elements

6d-series / 4th series / 7th period

	Acha	R	D	Sharma	Book 4th Series Has		Maths	Difficult	Range	Concepts
	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn
Z	89	104	105	106	107	108	109	110	111	112
7s	2	2	2	2	2	2	2	2	1	2
6d	1	2	3	4	5	6	7	8	10	10

Actinoids [90-103]

3d-Series			4d-Series			5d-Series		
Atomic number	Element	Electronic configuration	Atomic number	Element	Electronic configuration	Atomic number	Element	Electronic configuration
21	Sc	$[Ar]3d^1 4s^2$	39	Y	$[Kr]4d^1 5s^2$	57	La	$[Xe]5d^1 6s^2$
22	Ti	$[Ar]3d^2 4s^2$	40	Zr	$[Kr]4d^2 5s^2$	72	Hf	$[Xe]4f^{14} 5d^2 6s^2$
23	V	$[Ar]3d^3 4s^2$	41	Nb	$[Kr]4d^4 5s^1$	73	Ta	$[Xe]4f^{14} 5d^3 6s^2$
24	Cr	$[Ar]3d^5 4s^1$	42	Mo	$[Kr]4d^5 5s^1$	74	W	$[Xe]4f^{14} 5d^4 6s^2$
25	Mn	$[Ar]3d^5 4s^2$	43	Tc	$[Kr]4d^5 5s^2$	75	Re	$[Xe]4f^{14} 5d^5 6s^2$
26	Fe	$[Ar]3d^6 4s^2$	44	Ru	$[Kr]4d^7 5s^1$	76	Os	$[Xe]4f^{14} 5d^6 6s^2$
27	Co	$[Ar]3d^7 4s^2$	45	Rh	$[Kr]4d^8 5s^1$	77	Ir	$[Xe]4f^{14} 5d^7 6s^2$
28	Ni	$[Ar]3d^8 4s^2$	46	Pd	$[Kr]4d^{10} 5s^0$	78	Pt	$[Xe]4f^{14} 5d^9 6s^1$
29	Cu	$[Ar]3d^{10} 4s^1$	47	Ag	$[Kr]4d^{10} 5s^1$	79	Au	$[Xe]4f^{14} 5d^{10} 6s^1$
30	Zn	$[Ar]3d^{10} 4s^2$	48	Cd	$[Kr]4d^{10} 5s^2$	80	Hg	$[Xe]4f^{14} 5d^{10} 6s^2$

QUESTION

Write the electronic configuration of given atoms/ions:

i. Cr $[Ar]^{18}\,3d^5\,4s^1$

ii. Mn^{3+} $[Ar]^{18}\,3d^4$

iii. Fe^{3+} $[Ar]^{18}\,3d^5$

iv. Pd $[Kr]^{36}\,4d^{10}\,5s^0$

v. Pd^{2+} $[Kr]^{36}\,4d^8$

vi. Cu^{+} $[Ar]^{18}\,3d^{10}$

$Mn(25) \rightarrow [Ar]^{18}\,3d^5\,4s^2$

$Fe(26) \rightarrow [Ar]^{18}\,3d^6\,4s^2$

$Cu(29) \rightarrow [Ar]^{18}\,3d^{10}\,4s^1$



General Properties of Transition Elements (d-Block)

Physical Properties :

- All are metals
- Malleable and ductile (except mercury → liquid) at RT
- High thermal and electrical conductivity
- High tensile strength
- Metallic Lustre and sonorous

Malleable : Beaten into thin sheets

Ductile : Drawn into thin wires

Lustrous : Shiny / Chamak

Sonorous : Ringing sound

With the exceptions of Zn, Cd, Hg and Mn they have one or more typical metallic structures at normal temperatures.

General Properties of Transition

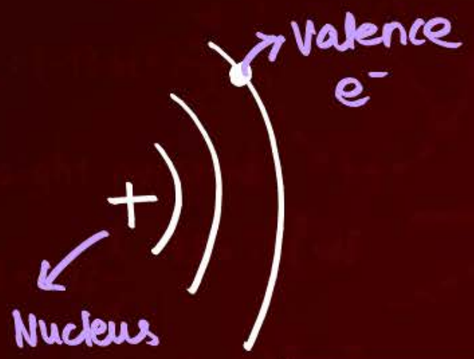
2. Atomic Radii :

On Moving left to right in a period, atomic radii decreases.

Size: s-Block > d– Block > p- Block

Hi! Tigers...

Valence electrons



Attractive Force

AF↑, size↓

$Z_{eff} = Z - \sigma$

- Z_{eff}: Effective Nuclear charge
- Z: Nuclear charge
- σ: Shielding Const

Repulsive Force

RF↑, size↑

Screening / shielding effect

Repulsion provided by the inner electrons on valence electrons.

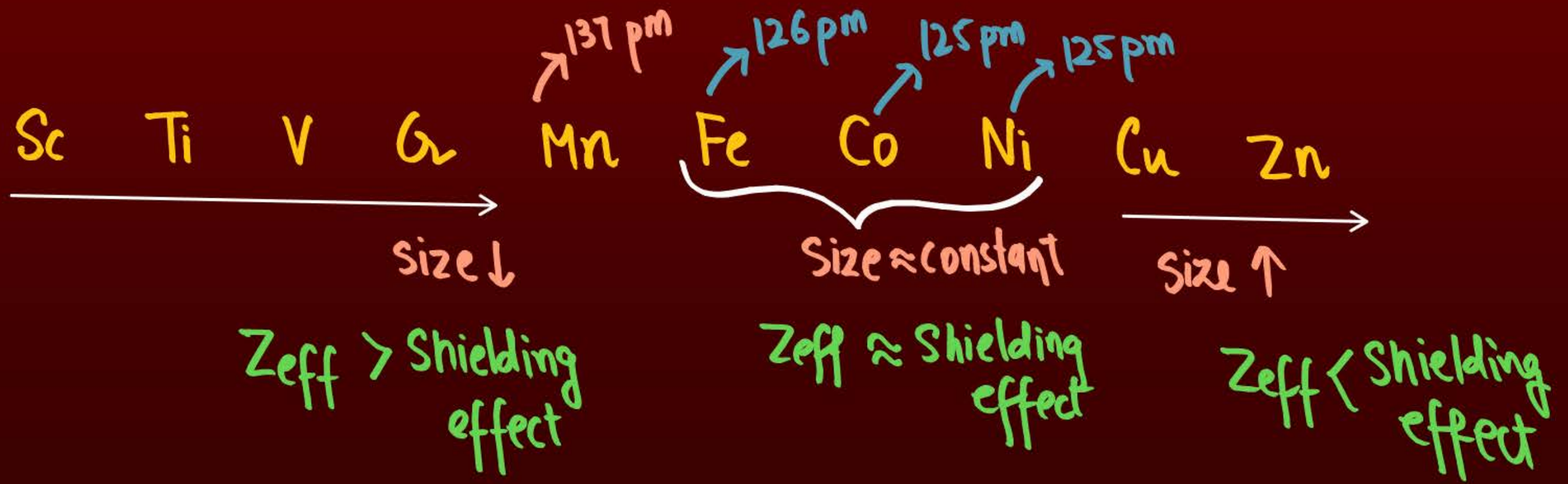
s > p > d > f

General Properties of Transition

2. Atomic Radii :

In 3d series (Sc – Zn)

On moving left to right , as atomic number increases , size first decreases up to middle , then becomes constant and then increases.



General Properties of Transition

2. **Atomic Radii for 3d series:**

$$Sc > Ti > V > Cr < Mn > Fe \approx Co \approx Ni < Cu < Zn$$

General Properties of Transition

Atomic Radii :

3d → 4d → 5d : size should increase, but 3d < 4d ≈ 5d (Except Group 3/ III B)

↳ Due to Lanthanoid Contraction

Gp-3	Gp - 4	Gp -12
3d $_{21}Sc$	$_{22}Ti$	$_{30}Zn$
Size ↑	Size ↑	
4d $_{39}Y$	$_{40}Zr$	$_{48}Cd$
Size ↑	160 pm	
5d $_{57}La$	$_{72}Hf$	$_{80}Hg$
	159 pm	

↳ Lanthanoids (4f) → poor screening

General Properties of Transition

3d → 4d → 5d :

Size should ↑se due to introduction of new principal shell.

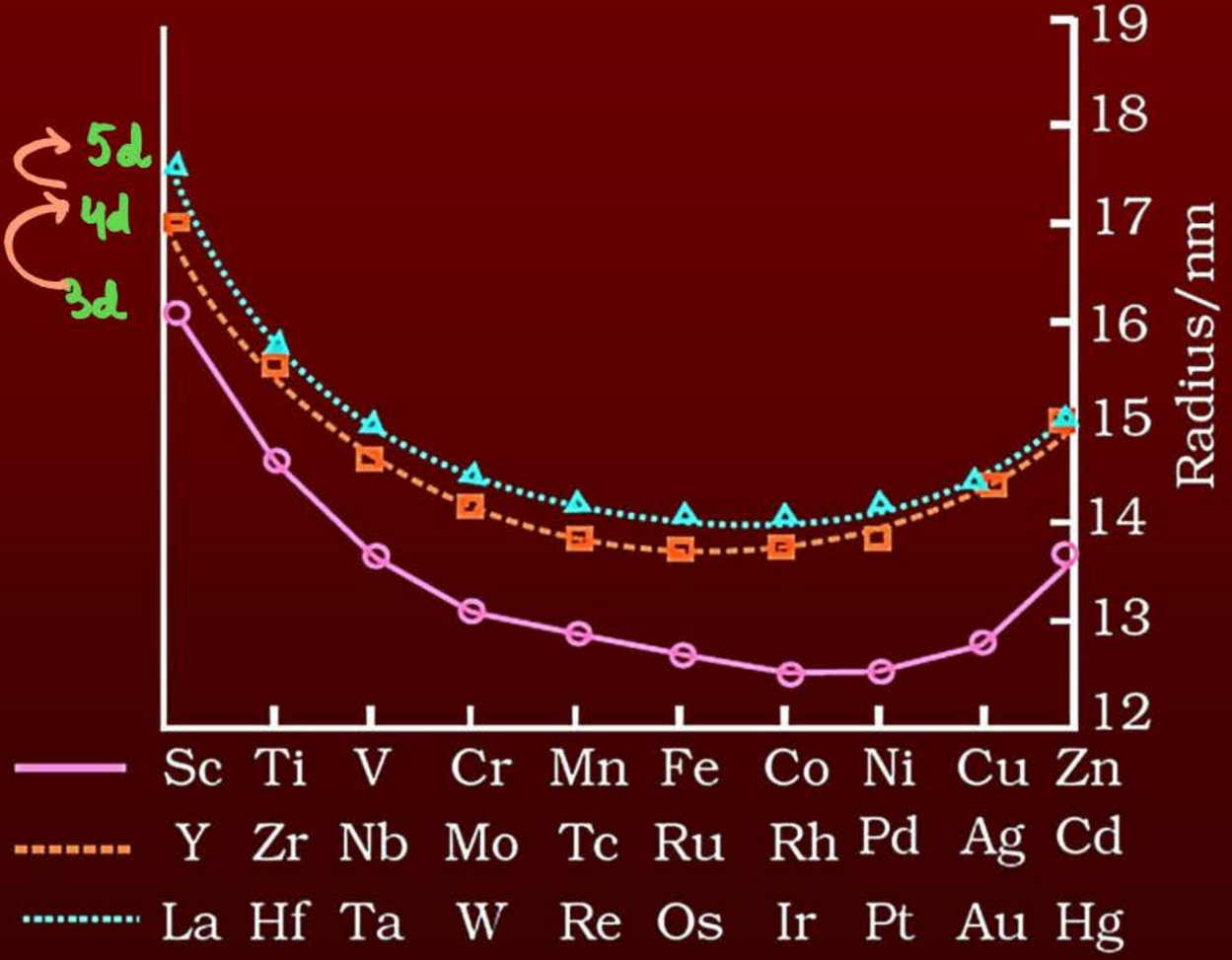
Note: The size of 4d elements is almost the same size of the 5d series elements.

This is due to filling of 4f before 5d orbitals which results in regular decrease in atomic radii & called as Lanthanoid Contraction e.g. (Zr 160 pm, Hf 159 pm)

General Properties of Transition

3d → 4d → 5d :





General Properties of Transition

Ionic Radii:

Size

- **↓se with ↑se in oxidation state.**

$Zn > Zn^{2+}$

$Mn^{2+} > Mn^{4+}$

$Fe^{2+} > Fe^{3+}$

- **In general, ions of the same charge in given series show progressive ↓se in radius with ↑sing atomic no.**

$_{21}Sc^{2+}$ $_{22}Ti^{2+}$ $_{23}V^{2+}$ $_{24}Cr^{2+}$ $_{25}Mn^{2+}$ $_{26}Fe^{2+}$ $_{27}Co^{2+}$ $_{28}Ni^{2+}$ $_{29}Cu^{2+}$ $_{30}Zn^{2+}$

Ionic Radii ↓ses as At. no. ↑ses

General Properties of Transition

Ionization Enthalpy : $\uparrow I.E \propto \frac{1}{\text{Size}} \downarrow$

Sg

- **From Left to right along the series, I.E. ↑se.** (Monichahie)

  (However, irregular trend in the 1st I.E. of 3d metals is due to irregularity in electronic configuration (E.C.) of 4s and 3d orbitals)

3d → 1st I.E

" Irregularity in size "

" Irregularity in Electronic configuration "

Homework

→ DPP Solve

→ Q's



THANK YOU